Quantum-Evolved Motion Priors in Particle-Based Missile Guidance

Soumyadip Chakrabarti
School of Computer Science and Engineering
Vellore Institute of Technology, Chennai
Tamil Nadu, India
soumyadip.2023@vitstudent.ac.in

Helen Vijitha P
School of Computer Science and Engineering
Vellore Institute of Technology, Chennai
Tamil Nadu, India
helen.vijitha@vit.ac.in

Jai Vinita L
School of Computer Science and Engineering
Vellore Institute of Technology, Chennai
Tamil Nadu, India
jaivinita.l@vit.ac.in

***Abstract*—Modern fighter planes still rely on their physical abilities during aerial engagements. The combination of high-G evasive maneuvers together with coordinated decoy deployment and active electronic countermeasures enables targets to evade classic PN motion models in currently used in missiles. Traditional state estimation methods that depend on fixed, predefined motion models and corrective routines tend to lose accuracy when targets suddenly change their behavior or move in unexpected, nonlinear ways that the models did not account for, in the long run the accuracy degrades exponentially until eventually the system fails to achieve interception at all. This research proposes a guidance system which maintains its Bayesian structure through its particle filter design while using neuroevolution based quantum components for prediction. The QNEAT-evolved circuit works like a biasing tool that steers particle movement toward known motion patterns while blocking updates that don't make sense. The process begins with the posterior belief, which is then turned into a discrete trajectory optimization problem that QAOA solves to find a new intercept path that follows engagement rules. Guidance updates are sent through tactical relays when the link is available, but if electronic interference cuts communication, the missile switches to autonomous control and uses terminal proportional navigation to keep tracking. Tests with synthetic engagement scenarios show that this system provides more stable interception than classical guidance methods under moderate maneuvering, while also staying reliable even when datalink connections are lost.**

***Index Terms*—Missile guidance architecture, Particle filtering; Quantum-evolved prediction, QAOA optimization, Electronic countermeasures, Proportional navigation.**

I. Introduction

Modern missile guidance struggles in contested airspace. High-G maneuvers, spectrally matched decoys, and coordinated ECM expose a core weakness: fixed motion models cannot adapt fast enough. Estimation error accumulates. Interception fails.

Classical filters assume linear dynamics [1]:

$$s_{t+1} = As_t + Bu_t + w_t \tag{1}$$

EKF and UKF relax this partially, but both depend on local linearization and predefined kinematics [2]. Under aggressive maneuvering, approximation error compounds through the guidance loop. Particle filters handle non-Gaussian dynamics, but still propagate using simplified priors: reactive, not anticipatory.

The main contributions of this paper are:

- A topology-evolved quantum predictive engine that restructures particle propagation via short-horizon state increments while preserving Bayesian weights.
- QAOA-based mid-course re-optimization reformulated as a constrained discrete trajectory selection problem.
- Three-tier separation: predictive intelligence (Tier 1), communication resilience (Tier 2), deterministic stabilization (Tier 3), with degradation logic for datalink denial.
- Full QNEAT specification: genome encoding, gate set $\mathcal{G} = \{H, X, \text{CNOT}\}$, fitness function, and training procedure for reproducibility.
- Quantitative validation across three maneuver profiles demonstrating consistent miss distance reduction over classical particle filtering.

Section II reviews related work. Section III presents the architecture. Section IV covers operational workflow. Section V evaluates simulation results. Section VI discusses structural implications, followed by conclusions in Section VII.

II. Literature Review of Contemporary Guidance Systems

Classical guidance laws such as Proportional Navigation (PN) react only to instantaneous kinematics. Against sustained curvature or abrupt nonstationary motion, this reactive gap grows miss distance increases, effectiveness drops. Gain adjustment is the usual fix [3], [4], but it is not predictive restructuring. Coordinated decoys and multi-agent deception make this worse, distorting interception geometry when guidance depends on fixed PN logic [5], [6].

Kalman-based estimators underpin most architectures [1], [2]. EKF linearizes locally; UKF samples deterministically. Both degrade under aggressive maneuvering coordinated turns, sinusoidal evasions, rapid climbs. Errors compound, guidance loops break down. Motion models stay simplistic by necessity: constant velocity, constant turn. Real targets do not cooperate. Reinforcement learning and deep networks have been tried, but dataset demands, generalization failures, and computational cost make them impractical for real-time loops. Trajectory optimization [7] and evolutionary UCAV strategies [8] show value at the aircraft planning level, but neither is embedded in interceptor mid-course guidance.

Particle filtering handles nonlinear, non-Gaussian dynamics well [9]. Yet standard PF still uses simple motion priors no curvature anticipation. Under high acceleration, weights collapse and degeneracy follows [10], [11]. Robust weighting helps against noise [12], but deception and ECM are structurally different problems. Deception-aware and consensus-based tracking approaches [13], [14] highlight the ceiling of memoryless, single-platform propagation under active interference.

Datalink dependence adds another failure mode. ECM degrades or severs midcourse communication links, leaving the missile on onboard estimators and reactive laws alone [15], [16]. Without a continuity mechanism that blends upstream prediction with local autonomy, guidance remains vulnerable. The gap is not estimation accuracy it is the absence of prediction-aware propagation that adapts structurally to maneuver geometry before curvature is fully expressed.

III. Proposed Multi-Tiered Guidance Architecture

We propose a hierarchical guidance framework distributing computation across three operational tiers. Rather than embedding all intelligence onboard, strategic prediction, tactical relay, and terminal autonomy are separated by design.

The overall structure is illustrated in Fig. 1.

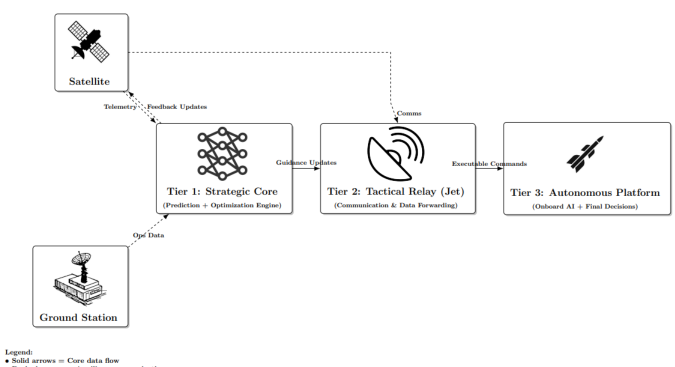

Fig. 1. Multi-Tiered Guidance Architecture showing Strategic Core, Tactical Relay, and Autonomous Platform.

A. Tier 1 – Strategic Core (Prediction and Optimization Engine)

Tier 1 performs offboard strategic optimization and advisory generation. A classical particle filter produces feasible target state distributions under nonlinear dynamics. QNEAT reshapes the prediction step: evolved discrete-gate circuits bias short-horizon propagation toward learned curvature patterns without altering posterior weights or Sequential Monte Carlo consistency. The refined hypothesis set feeds a QAOA-based re-optimization layer that selects kinematically feasible guidance advisories. Outputs are constrained directional and velocity updates for mid-course correction. This tier is computationally heavy but not latency-critical.

1) QNEAT Framework: QNEAT evolves discrete quantum circuit topologies to approximate short-horizon motion increments. No gradient-based training is used. Each genome encodes gate genes:

$$g = (\text{innovation\_id},\ \text{gate\_type},\ \text{target\_qubits}), \tag{2}$$

with gate types drawn from the fixed set:

$$\mathcal{G} = \{H,\ X,\ \text{CNOT}\}. \tag{3}$$

The circuit acts on $n_q = 4$ qubits over $S = 256$ measurement shots. The empirical mean output for qubit j is

$$\bar{b}_j = \frac{1}{S}\sum_{s=1}^{S} b_j(s), \quad y_j = (\bar{b}_j - 0.5)\,c, \tag{4}$$

giving motion increment $\Delta x_k^{(i)} = Q(\phi_k)$, which replaces the fixed kinematic transition:

$$x_{k+1}^{(i)} = x_k^{(i)} + \Delta x_k^{(i)} + w_k^{(i)}. \tag{5}$$

Bayesian weights and measurement updates remain unchanged. All evaluations run on classical simulation; no quantum speedup is claimed. Fitness is:

$$e_m = \alpha\|a_{\text{pred}} - a_{\text{true}}\| + (1-\alpha)\|\Delta v_{\text{pred}} - \Delta v_{\text{true}}\|, \tag{6}$$

where M is the number of training samples. The mean prediction error across all samples is

$$\bar{e} = \frac{1}{M}\sum_{m=1}^{M} e_m, \tag{7}$$

and fitness is defined as

$$F = \frac{1}{1+\bar{e}}. \tag{8}$$

Speciation uses compatibility distance $\delta = c_1 E/N + c_2 D/N$. The full training loop is given in Algorithm 1.

Algorithm 1 QNEAT Training Procedure

Require: Training dataset D, population size P, generations G

Ensure: Best-performing genome

Initialize population of random circuit genomes
for $g = 1$ to G **do**
    for each genome **do**
        Construct quantum circuit from its gene sequence
        Evaluate predictions over training dataset D
        Compute fitness $F = 1/(1 + \bar{e})$
    end for
    Group genomes into species by compatibility distance δ
    Preserve elite genomes; generate offspring via crossover and mutation
    Replace previous generation
end for
return highest-fitness genome

2) QAOA Re-Optimization Layer: QAOA runs entirely in Tier 1 offboard, on the strategic node. The missile never executes QAOA directly. Pre-packaged guidance vectors are relayed before they are needed. QAOA solves a discrete trajectory feasibility problem under time-to-go, closing velocity, and acceleration constraints. If no update arrives, the missile holds its last valid vector and transitions to autonomous PN.

B. Tier 2 – Tactical Relay (Jet Platform)

Tier 2 bridges strategic computation and the interceptor via the launch aircraft or a cooperating platform. It receives guidance updates from Tier 1, performs integrity checks, and forwards command packets to the missile. Under ECM, it manages retransmission, encryption, and timing synchronization. If link quality drops below threshold, it signals Tier 3 to assume autonomy. This tier decouples high-level optimization from missile bandwidth constraints.

C. Tier 3 – Autonomous Platform (Missile Onboard AI)

Tier 3 handles onboard sensing, filtering, and terminal guidance. While the datalink holds, the missile executes upstream corrections. When communication is lost, it transitions to autonomous pursuit: onboard nonlinear filtering continues, local stabilization maintains course, and terminal PN governs final correction. The last validated guidance update is retained as directional bias. At close range, classical PN is preferred for its geometric robustness and low computational overhead.

IV. Operational Workflow

The guidance system operates as a staged decision pipeline. Each tier contributes at a different temporal and computational scale across four phases.

A. Initialization and Target State Construction

At launch, the Strategic Core constructs initial target state estimates from radar tracks, telemetry feeds, and distributed

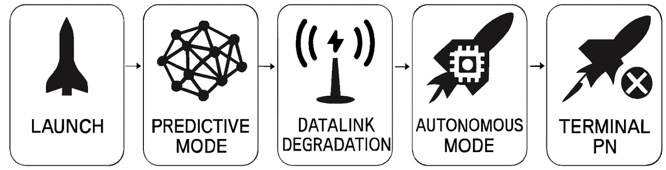

Fig. 2. Mode Transition Diagram

sensor inputs. A particle filter builds a probabilistic state representation, with the prediction stage augmented by the QNEAT-evolved motion prior. Short-horizon increments are refined without altering Bayesian updates or weight computation. The output is a constrained belief distribution over plausible near-future target states, not a single trajectory.

B. Mid-Course Predictive Refinement

The QNEAT circuit biases particle propagation toward motion patterns observed during training. Increments are clipped via affine scaling to prevent unrealistic state jumps, and conditioned on belief-derived features ϕ_k to ensure proposed states stay within regions of non-negligible likelihood before the Bayesian update. Candidate trajectories are then evaluated under time-to-go feasibility, closing velocity, and acceleration limits. QAOA selects directionally coherent advisories without full rollouts and forwards them as guidance vectors to the Tactical Relay.

C. Tactical Relay and Communication Handling

The relay forwards guidance vectors to the missile over secure datalink channels while maintaining communication state awareness. Under ECM, it performs retransmission attempts, switches modulation or frequency profiles, and signals degraded link state. Temporary disruptions do not immediately terminate predictive correction. If connectivity drops below threshold, a mode transition signal is issued to Tier 3.

D. Autonomous Mode Transition

Communication loss does not collapse the system. The last validated guidance update is retained as directional bias, onboard nonlinear filtering continues state estimation, and classical PN governs terminal correction using LOS rate and closing velocity. At short range, geometric guidance outperforms predictive computation due to lower latency. Guidance shifts from distributed prediction to localized stabilization as range decreases.

E. Terminal Phase

Predictive quantum refinement is disabled at terminal range to prevent overfitting stale information and ensure deterministic behavior. Control authority is governed by instantaneous LOS rate, closing velocity, and acceleration limits. Control transitions sequentially from strategic prediction through relay-assisted correction and autonomous stabilization to deterministic terminal PN.

F. Resilience Mechanisms

Three mechanisms guard against ECM-induced failure.

1) Predictive Buffering: Updates are transmitted as incremental advisories, allowing onboard buffering. Under link loss, the missile retains its most recent directional bias rather than reverting to purely reactive tracking.

2) Tiered Autonomy: The onboard filtering pipeline and classical guidance law keep the missile operational under complete communication denial, ensuring independent estimation, stable geometric convergence, and reduced jamming vulnerability.

3) Graceful Degradation: The architecture transitions smoothly from full predictive guidance to reduced updates to pure autonomous pursuit. In contested airspace, intermittent disruption is far more common than total link loss. This structured fallback ensures continuity under all degradation levels.

V. Simulation Framework and Observations

A. Simulation Setup

A controlled simulation environment was built to assess structural behavior of the multi-tiered architecture under nonlinear engagement conditions. QNEAT circuit evolution was performed entirely offline; the evolved discrete-gate circuit replaced the deterministic transition model within the SMC prediction stage, leaving Bayesian weight updates unchanged. All runs used identical missile dynamics, sensing models, and acceleration constraints. Only the prediction mechanism differed between baseline and proposed configurations.

Target profiles covered three maneuver types: sustained coordinated turns, oscillatory lateral evasions, and compound multi-phase maneuvers combining lateral and vertical acceleration. These profiles expose memoryless propagation weaknesses and mirror conditions studied in evasive maneuver optimization [8], [17], with focus shifted to interceptor-side restructuring rather than target-side synthesis [16]. State estimation follows:

$$x_{k+1}^{(i)} = x_k^{(i)} + \Delta x_k^{(i)} + w_k^{(i)} \tag{9}$$

where $\Delta x_k^{(i)}$ is refined by the quantum predictor in the proposed configuration and fixed kinematic assumptions in the baseline.

TABLE I
Simulation and Algorithm Parameters

Parameter	Value
Particle count (N)	500
QNEAT gate set ($\mathcal{G}$)	$\{H, X, \text{CNOT}\}$
Qubits (n_q)	4
Measurement shots (S)	256
Evolution generations (G)	200
Population size (P)	100
Fitness weight (α)	0.5
Branching horizon (H)	3 steps
Branch pruning threshold (τ)	0.01
Integration timestep (Δt)	0.05 s
Simulation mode	Classical (no quantum hardware)

B. Quantitative Results

Terminal interception performance across all three maneuver profiles is summarised in Table II.

TABLE II
Terminal Miss Distance and Mean Estimation Error Across Maneuver Profiles

Maneuver	CTPF Miss (m)	QCTPF Miss (m)	CTPF Mean Err (m)	QCTPF Mean Err (m)
Oscillatory + Vertical	7589.62	**1331.05**	355.77	352.51
Bidirectional Turn	8890.28	**515.04**	32.67	23.56
Compound Multi-Phase	14367.68	**5950.24**	1784.76	1141.01

QCTPF reduced terminal miss distance across the evaluated maneuver profiles, particularly under rapidly changing nonlinear geometries. The gains reflect earlier curvature adaptation rather than lower instantaneous estimation error mean errors remain comparable between CTPF and QCTPF, confirming that predictive belief restructuring drives closed-loop improvement, not noise reduction.

Table III benchmarks all methods under identical coordinated-turn dynamics, sensing conditions, and PN guidance laws.

TABLE III
Benchmark Comparison Against Classical Nonlinear Estimators

Method	Mean Position Error (m)	Miss Distance (m)
EKF	5821.26	6130.39
UKF	5805.66	6190.33
CTPF	1301.78	4085.22
QCTPF	1527.00	4481.78

EKF and UKF show substantially larger miss distances, reflecting the limits of linearized Gaussian estimation under nonlinear dynamics. Particle-based methods outperform both. QCTPF does not surpass the optimized CTPF in this standardized benchmark—coordinated-turn dynamics favor the classical prior—but it demonstrates feasible integration of quantum-evolved prediction within an SMC architecture under identical constraints. The maneuver-specific results in Table II better reflect QCTPF's advantage under profile-dependent nonlinear geometries.

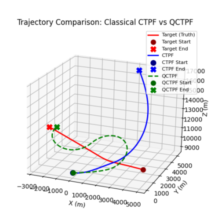

Fig. 3. Oscillatory + vertical maneuver: QCTPF vs. classical PF trajectory comparison.

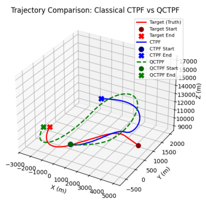

Fig. 4. Bidirectional turn maneuver: trajectory comparison under curvature reversal.

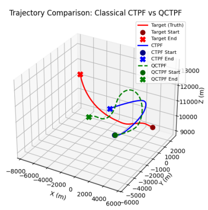

Fig. 5. Compound multi-phase maneuver: terminal convergence under lateral and vertical transitions.

C. Mid-Course Behavior

Across maneuver profiles, distinct structural differences emerged between classical and proposed architectures.

In sustained coordinated turns, the classical setup showed low step-wise estimation error but increasing terminal miss distance, reflecting reactive lag that failed to anticipate curvature. The proposed architecture, by contrast, achieved better mid-course alignment: the quantum-evolved prior biased short-horizon propagation toward curvature-consistent increments, keeping guidance updates aligned with evolving motion. The gain appeared less in mean error reduction and more in improved geometric convergence.

In classical PF, a maneuver reversal only influences particles after the measurement update registers the shift, introducing an inherent lag. QNEAT conditions propagation on kinematic features ϕ_k encoding angular rate and closing geometry, anticipating heading change before measurement confirms it. This explains the gap between mean estimation error and terminal miss distance in Table II: instantaneous accuracy stays comparable, but earlier belief alignment gives the guidance law more time to correct.

D. Oscillatory and Multi-Phase Maneuvers

Oscillatory lateral evasions forced rapid reversals, where classical particle propagation faltered degeneracy and model mismatch drove terminal miss distances upward whenever curvature shifted abruptly. The multi-tiered design proved more resilient, refining increments and re-evaluating feasible paths without altering Bayesian weights, thereby cutting miss distance. Under compound multi-phase motion with lateral acceleration, oscillation, and vertical transitions, both systems saw error growth, yet the proposed framework consistently held lower terminal miss distance under identical sensing. Even imperfect short-horizon prediction added stability, tightening closed-loop convergence.

E. Communication Loss Emulation

Although explicit ECM modeling was not implemented, controlled simulations were conducted where mid-course updates were intentionally halted during engagement.

When communication was discontinued, the missile transitioned to autonomous mode and continued pursuit using onboard filtering and proportional navigation. The structured fallback preserved engagement continuity and avoided catastrophic divergence.

While this does not constitute a full ECM resilience validation, it demonstrates that the architecture can maintain operational stability under communication interruption.

F. Structural Comparison with Existing Guidance Frameworks

PN-GAPF [18] improves particle diversity through genetic resampling but leaves the motion prior unchanged. Our architecture intervenes at propagation—a QNEAT-evolved circuit replaces the deterministic transition model while Bayesian weights remain intact.

Wang et al.'s pseudospectral MPC [7] solves a full nonlinear optimal control problem via Hamiltonian analysis. Effective, but deterministic and computationally heavy. We generate short-horizon anticipatory corrections within particle-based uncertainty instead, reverting to classical PN at terminal range. Segmented cooperative guidance [19] similarly concedes that fixed-structure laws fail under high-G transitions by partitioning engagements into phases.

Learning-based approaches [20] confirm adaptive strategies outperform fixed priors under aggressive evasion. Distributed estimation under jamming [21] directly motivates the relay tier. Nonlinear 3D formulations [22] improve realism at high solver cost. The distinction here is conceptual: adaptivity is embedded in belief propagation, not delegated to a deterministic solver.

VI. Computational Feasibility

QNEAT evolution runs entirely offline, contributing zero onboard burden during engagement. At runtime, the missile performs only forward circuit evaluation, bounded branching, and terminal PN, all lightweight operations.

Offline training complexity scales as:

$$O(G \cdot P \cdot M \cdot L \cdot S) \tag{10}$$

where G is generations, P population size, M training samples, L average circuit depth, and S measurement shots. Online cost stays bounded through short prediction horizons, low-likelihood branch pruning, and relay-assisted computation in Tier 1. Heavy predictive restructuring and QAOA refinement remain offboard; Tier 3 handles only local filtering and deterministic PN fallback.

All evaluations used classical simulation. Future deployment could leverage FPGA acceleration or hybrid quantum-classical embedded systems.

TABLE IV
COMPARISON OF REPRESENTATIVE GUIDANCE FRAMEWORKS

Authors	Paper Title	Year	Framework Type	Core Characteristics
Liu et al. [18]	Tracking Air-to-Air Missile Using Proportional Navigation Model with Genetic Algorithm Particle Filter	2016	PN-GAPF (Particle Filter + GA)	Fixed PN motion model; genetic resampling for particle diversity; prediction model unchanged
Wang et al. [7]	Optimal Midcourse Guidance Using Linear Gauss Pseudospectral MPC	2025	Optimal Control (LGPMPC)	Hamiltonian-based nonlinear optimal control; boundary value problem solution; deterministic trajectory optimization
Fan et al. [19]	Time-Cooperative Guidance Under Evasive Maneuvering	2024	Segmented Cooperative Guidance	Trajectory partitioning under evasive motion; phase-based guidance refinement
Schneider et al. [20]	Autonomous Evasive Maneuvers via Reinforcement Learning	2025	Learning-Based Guidance	Deep RL-based adaptive motion strategies; policy learning for nonlinear engagements
Papaioannou et al. [21]	Distributed Estimation and Control Under Jamming	2023	Distributed Robust Estimation	Consensus-based tracking under communication degradation; resilience to jamming
Samrat et al. [22]	Three-Dimensional Nonlinear Guidance with FOV Constraints	2025	Nonlinear Deterministic Guidance	Constraint-aware 3D nonlinear guidance; computationally intensive solver
Proposed Work	Quantum-Evolved Motion Priors in Particle-Based Missile Guidance		Hierarchical Predictive Architecture	Topology-evolved motion prior; prediction-stage modification; Bayesian consistency preserved; relay-tier resilience; PN terminal fallback

VII. CONCLUSION

This work presents a multi-tiered missile guidance architecture that replaces the fixed particle filter motion prior with a QNEAT-evolved discrete-gate predictor, offloads trajectory re-optimization to a QAOA-based strategic node, and maintains autonomous PN fallback when the datalink fails. Against classical Gaussian estimators, the architecture shows improved interception stability across oscillatory, bidirectional, and compound multi-phase maneuver profiles. No quantum speedup is claimed; all gains stem from predictive restructuring and tiered design. Future work will extend this toward stability guarantees under adversarial maneuvers and hybrid quantum-classical embedded deployment.

REFERENCES

[1] C. Montella, "The kalman filter and related algorithms: A literature review," U.S. Air Force Institute of Technology, Tech. Rep., 2011.

[2] E. A. Wan and R. van der Merwe, "The unscented kalman filter for nonlinear estimation," in *Proceedings of the IEEE Adaptive Systems for Signal Processing, Communications, and Control Symposium*, 2000, pp. 153–158.

[3] P. Zarchan, *Tactical and Strategic Missile Guidance*, 7th ed. Reston, VA: American Institute of Aeronautics and Astronautics (AIAA), 2012.

[4] N. A. Shneydor, *Missile Guidance and Pursuit: Kinematics, Dynamics and Control*. Chichester, UK: Horwood Publishing, 1998.

[5] E. Bildik, A. Tsourdos, A. Perrusquía, and G. İnalhan, "Decoys deployment for missile interception: A multi-agent reinforcement learning approach," *Aerospace*, vol. 11, no. 8, p. 684, 2024. [Online]. Available: https://doi.org/10.3390/aerospace11080684

[6] E. Bildik and A. Tsourdos, "Clustering and cooperative guidance of multiple decoys for defending a naval platform against salvo threats," *Aerospace*, vol. 11, no. 10, p. 799, 2024. [Online]. Available: https://doi.org/10.3390/aerospace11100799

[7] J. WANG, W. CHEN, L. YANG, and X. GONG, "An optimal midcourse guidance method for dual pulse air-to-air missiles using linear gauss pseudospectral model predictive control method," *Chinese Journal of Aeronautics*, vol. 38, no. 2, p. 103199, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1000936124003364

[8] Z. Yang, D. Zhou, H. Piao, K. Zhang, W. Kong, and Q. Pan, "Evasive maneuver strategy for ucav in beyond-visual-range air combat based on hierarchical multi-objective evolutionary algorithm," *IEEE Access*, vol. 8, pp. 46 605–46 623, 2020.

[9] A. Doucet, N. de Freitas, and N. Gordon, Eds., *Sequential Monte Carlo Methods in Practice*, ser. Springer Series in Statistics. Springer, 2001.

[10] C. Snyder, T. Bengtsson, P. Bickel, and J. Anderson, "Obstacles to high-dimensional particle filtering," *Monthly Weather Review*, vol. 136, no. 12, pp. 4629–4640, 2008.

[11] J. D. Hol, T. B. Schön, and F. Gustafsson, "On resampling algorithms for particle filters," *IEEE Signal Processing Magazine*, vol. 23, no. 5, pp. 140–148, 2006.

[12] Q. Wu, L. Chen, Y. Li, Z. Wang, S. Yao, and H. Li, "Reweighted robust particle filtering approach for target tracking in automotive radar application," *Remote Sensing*, vol. 14, no. 21, p. 5477, 2022. [Online]. Available: https://doi.org/10.3390/rs14215477

[13] A. Javadpour *et al.*, "A comprehensive survey on cyber deception techniques and countermeasures," *Journal of Information Security and Applications*, vol. 74, p. 103350, 2024. [Online]. Available: https://doi.org/10.1016/j.jisa.2024.103350

[14] Y. Yang, K. Da, Y. Zhu, S. Xiang, and Q. Fu, "Consensus-based target tracking against deception jamming in distributed radar networks," *IET Radar, Sonar & Navigation*, vol. 17, no. 4, pp. 683–700, 2023. [Online]. Available: https://doi.org/10.1049/rsn2.12371

[15] X. Li and S. Li, "Assessment of anti-jamming capabilities in infrared-guided missile systems," *Physica D: Nonlinear Phenomena*, vol. 481, p. 34737, 2025, early Access / In Press. [Online]. Available: https://doi.org/10.1016/j.physd.2025.34737

[16] Z. Tian, M. Danino, Y. Bar-Shalom, and B. Milgrom, "Estimation-based missile threat detection and evasion maneuver for a low-altitude aircraft," in *Signal Processing, Sensor/Information Fusion, and Target Recognition XXXII*, vol. 12547. SPIE, June 2023, pp. 27–35.

[17] G. Wang, X. Liu, and R. Yang, *Trajectory Optimization for Aircraft Evasive Maneuver Against a Missile by Convex Optimization*, 03 2025, pp. 371–383.

[18] H. Liu, L. Yu, C. Ruan, and Z. Zhou, "Tracking air-to-air missile using proportional navigation model with genetic algorithm particle filter," *Mathematical Problems in Engineering*, vol. 2016, no. 1, p. 3921608, 2016. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1155/2016/3921608

[19] Z. Fan, Y. Zhang, and H. Li, "Time-cooperative guidance strategy under evasive maneuvering conditions," *Journal of Marine Science and Engineering*, vol. 12, no. 6, p. 1018, 2024.

[20] M. Schneider, A. Keller, and D. Furfaro, "Autonomous evasive maneuvers against missiles through reinforcement learning," in *AIAA SciTech Forum*, 2025.

[21] S. Papaioannou, P. Kolios, and G. Ellinas, "Distributed estimation and control for jamming an aerial target with multiple agents," *IEEE Transactions on Mobile Computing*, vol. 22, no. 12, pp. 7203–7217, 2023.

[22] A. R. Samrat and P. Tsiotras, "Three-dimensional nonlinear guidance with field-of-view constraints," *IEEE Transactions on Aerospace and Electronic Systems*, 2025.